

Fig. 2: An actual-size PCB layout for the circuit

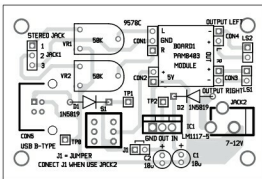


Fig. 3: Component layout for the PCB

within seconds.

Stereo audio output from an external source available at 3.5mm stereo input Jack1 is routed to PAM8403 module through two independent volume-control potentiometers VR1 and VR2. Similarly, 5V DC supply from USB port of the device available at USB input CON5 is extended to PAM8403 module through a protection diode (D1) and on/off switch (S1). The external DC input at Jack2 can be used to power the system from any 7V to 12V DC power source.

The LM1117-5 is a fixed 5V low-dropout-voltage regulator which offers current limiting and thermal shutdown. This IC is commonly available in TO-220 (regular) and SOT-223 (SMD) packages. Note that, a minimum of 10 μ F tantalum capacitor is required at the input and the output of LM1117-5 to improve transient response and stability.

Construction and testing

An actual-size, single-side PCB for the USB stereo audio amplifier is shown in

Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic box.

Connect CON1 through CON4 on Board1. Fix Jack1 on PCB for stereo audio input.

Fix USB B-type connector CON5 and

Jack2 on the PCB for the two power supplies. Fix switch S1 on front panel of the box for power supply and pots VR1 and VR2 for volume control. Connect jumper J1 in the PCB (see Fig. 3) if you are using the 7-12V supply via Jack2.

For connecting the speakers, two connectors are available on the PCB. Before connecting the load verify that the voltages at test points are as per table. ●

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